Quantum Black Holes and The Gravitational Path Integral

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Bekenstein's Formula

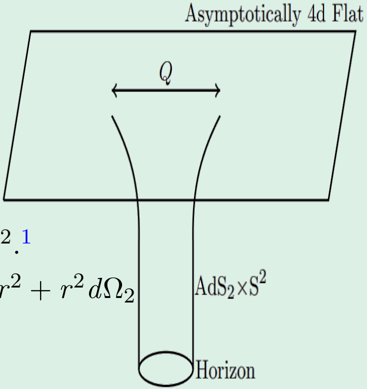
In classical general relativity, (Bekenstein 1973) proposed that the entropy of a black hole S_{BH} is proportional to the horizon's area, A

$$S_{BH} = \frac{A}{4}\frac{k}{\ell_p^2} \qquad \ell_p \equiv \sqrt{\frac{\hbar G}{c^3}}$$

where k is Boltzmann's constant and ℓ_p is the Planck length.

A Triumph of String theory

- A **microscopic** description was found in (Strominger and Vafa 1996) for the case of 5 d black holes $D1 - D5$ system.
- This was an important input into the discovery in 1997 of the infamous Anti-De Sitter/Conformal Field Theory correspondence.
- In this talk we will primarily discuss black holes in the context of four dimensional $\mathcal{N} = 8$ supergravity.



- The Near Horizon Region: $AdS_2 \times S^2$.[1]
- Metric: $ds^2 = -f(r)\,dt^2 + [f(r)]^{-1}\,dr^2 + r^2\,d\Omega_2$

where $f(r) := 1 - \frac{2M}{r} + \frac{q^2+p^2}{r^2}$

We have a BPS bound

$$M^2 = p^2 + q^2$$

[1] Image Credit: (Mertens, Turaci, 2022)

1/8 BPS Black Holes in String Theory

The Bekenstein-Hawking entropy of these black holes is given below (Kallosh and Kol 1996).

$$\boxed{S_{BH}\left(\Delta\right) = \pi\sqrt{\Delta\left(n,\ell\right)}}$$

$$\Delta = 4n - \ell^2 \qquad n \in \mathbb{N}, \ell \in \mathbb{Z}$$

where n and ℓ are determined by $\mathcal{Q}$.

We define the modular group as

$$SL\left(2,\mathbb{Z}\right) := \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \middle| a,b,c,d \in \mathbb{Z},\ ad-bc=1 \right\}.$$

For any $\gamma \in SL\left(2,\mathbb{Z}\right)$ and $\tau \in \mathbb{H}$, where $\mathbb{H} := \left\{\tau \in \mathbb{C} \middle| \Im\left(\tau\right) > 0\right\}$

$$f\left(\gamma\tau\right) = \left(c\tau + d\right)^k f\left(\tau\right).$$ [2]

[2] k is called the *weight* of the modular form.

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The **generators** of $SL\left(2,\mathbb{Z}\right)$ are

$$f\left(-\frac{1}{\tau}\right) = \tau^k f\left(z\right), \qquad f\left(z+1\right) = f\left(z\right).$$

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The Generators of $SL(2, \mathbb{Z})$

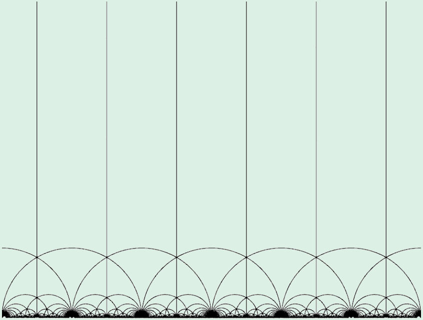

Figure (a): Action of T generator.

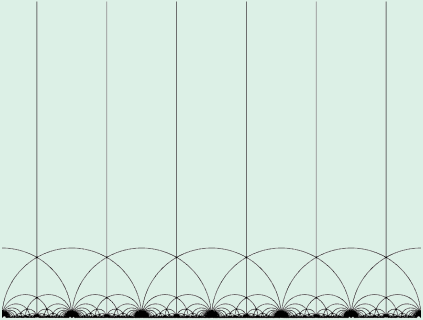

Figure (b): Action of S generator.[3]

[3] Adapted from gifs by (Nihargargava 2017).

Automorphic Forms in the Microscopic Interpretation

We can define

$$q := e^{2\pi i\tau},\ \zeta := e^{2\pi iz} \qquad \tau \in \mathbb{H},\ z \in \mathbb{C}.$$

Then the generating function of the degeneracy of BPS states is (Maldacena, Moore, and Strominger 1999)

$$\varphi_{-2,1}(\tau, z) = \frac{\vartheta_1(\tau, z)^2}{\eta^6(\tau)} = \sum_{n,\ell\in\mathbb{Z}} c_{-2,1}(n,\ell)\, q^n\zeta^\ell.$$

$$\vartheta_1(\tau, z) := \sum_{n\in\mathbb{Z}+\frac{1}{2}} (-1)^n q^{\frac{n^2}{2}}\zeta^n, \qquad \eta(\tau) := q^{\frac{1}{24}} \prod_{n=1}^{\infty} (1-q^n).$$

[4] We call m the index of the Jacobi Form.

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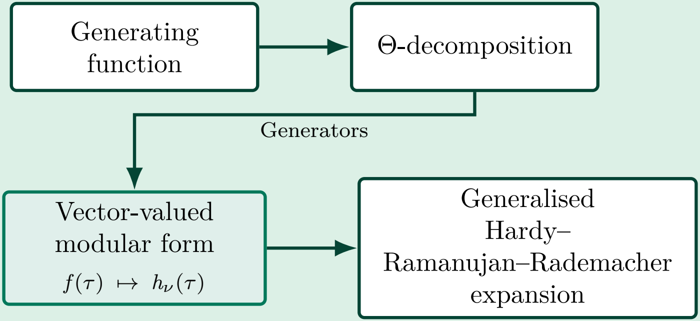
We have a theta expansion for ϑ_1.[4]

$$\varphi_{k,m}(\tau, z) = \sum_{\mu \bmod 2m} h_\mu \Theta_{m,\mu}(\tau, z) \qquad \mu \in \frac{\mathbb{Z}}{2m\mathbb{Z}}$$

[4] We call m the index of the Jacobi Form.

Microscopic Interpretation for $\mathcal{N} = 8$

Overview of the computation



[4] See Apostol (1990), Thm. 5.10.

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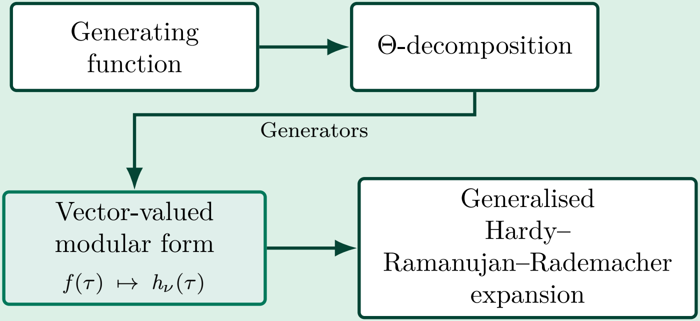

Vector-Valued Modular transformation:

$$h_\nu(\gamma) = (c\tau + d)^{\frac{1}{2}-k}\, M^{-1}_{\nu\mu}(\gamma)\, h^\mu(\gamma\tau)$$

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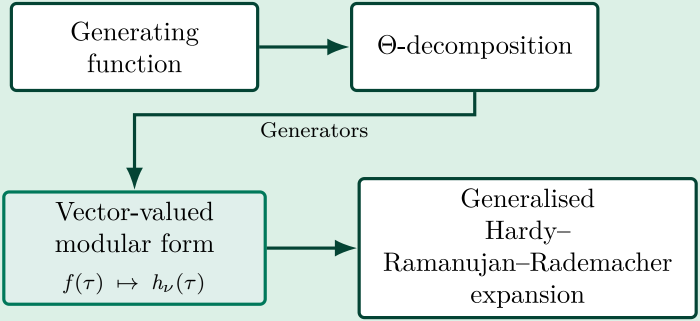

Vector-Valued Modular transformation:

$$h_\nu(\gamma) = (c\tau + d)^{\frac{1}{2}-k} M^{-1}_{\nu\mu}(\gamma)\, h^\mu(\gamma\tau)$$

Hardy–Ramanujan–Rademacher expansion:

$$C(\Delta) = \frac{1}{2\pi i} \oint_{|q|=r} \frac{f(\tau)}{q^{\frac{\Delta}{4}+1}}\, dq, \qquad f(\tau) \to h_\nu(\tau)$$

[4] See Apostol (1990), Thm. 5.10.

Microscopic Result for $\mathcal{N} = 8$

A replication of Bekenstein-Hawking.

$$\boxed{C(\Delta) = 2\pi \left(\frac{\pi}{2}\right)^{\frac{7}{2}} \sum_{c=1}^{\infty} c^{-\frac{9}{2}} K_c(\Delta) \tilde{\mathcal{I}}_{\frac{7}{2}} \left(\frac{\pi\sqrt{\Delta}}{c}\right).}$$

$$W_{\text{micro}}(\Delta) = (-1)^{\Delta+1} C(\Delta).$$

Appendix A

Taking the asymptotic expansion as $(\Delta \to \infty, \tilde{\mathcal{I}}_P \to e^{\frac{\pi\sqrt{\Delta}}{c}})$.

$$\boxed{S = \log W_{\text{micro}}(\Delta) \sim \pi\sqrt{\Delta} + \dots}$$

The Gravitational Path Integral

$$\mathcal{W}(q,p) = \int [\mathcal{D}g\mathcal{D}\Psi\mathcal{D}A\mathcal{D}\Phi]\, e^{-S^{\text{bulk}}_{\text{sugra}} - S^{\text{bdry}}_{\text{sugra}}}$$

- Fix the boundary conditions for the fields and background
- Functional integral which is **inherently infinite-dimensional**
- In supersymmetric settings, **localisation** can ultilised to reduce certain observables to a finite dimensional integral.

The Field Content of $\mathcal{N} = 8$ Supergravity

The fields in which we integrate over.

- **1 Graviton Multiplet**
- 6 Gravitino Multiplets
- **15 Vector Multiplets**
- 10 Hypermultiplets

This gives us 28 gauge fields and 70 scalars

In 4 d BPS black holes in $\mathcal{N} = 8$ supergravity the gravitational path integral reduces via equivariant localisation to the **quantum entropy function** (Dabholkar, Gomes, and Murthy 2011)

$$\mathcal{W}^c = \int \left[d^{n_\nu+1}\phi\right]_c \exp\left[-\frac{\pi q_\Lambda \phi^\Lambda}{c} + \frac{4\pi}{c}\Im F\left(\frac{\phi^\Lambda + ip^\Lambda}{2}\right)\right] \mathcal{Z}^{Q_{\mathrm{eq}}\mathcal{V}}_{\text{1-loop}} \mathcal{Z}_c^{\text{top}}$$

where $\Lambda = 0, \ldots, n_V$.

After localisation we obtain a contribution from the orbifold saddle points to get

$$W^c(q,p) = \sqrt{c} K_c(\Delta) \int d^{16}\phi \sqrt{\frac{-2C_{IJK}p^I p^J p^K \ \det\ (3C_{IJ})}{c^{16}\left(\phi^0\right)^{18}}} \frac{1}{c}\left(\frac{\phi^0}{-2C_{IJK}p^I p^J p^K}\right)^4$$

$$\cdot \exp\left[-\frac{\pi C_{IJK}p^I p^J p^K}{c\phi^0} + \frac{3\pi C_{IJK}\phi^I \phi^J p^K}{c\phi^0} - \frac{\pi q_0 \phi^0}{c} - \frac{\pi q_I \phi^I}{c}\right]$$

The agreement between the microscopic and macroscopic description is manifest.

$$\boxed{S = \log W_{\text{micro}}(\Delta) \sim \pi\sqrt{\Delta} + \ldots}$$

A Reduction of Supersymmetry

What is the impact of a reduction of Supersymmetry?

- Generating function of BPS degeneracy of states
- Automorphic Decomposition
- The prepotential
- One-loop
- The field content
- Multi-centered black hole contributions

The Microscopic Interpretation for $\mathcal{N} = 4$

$$d\left(m, n, \ell\right) = \left(-1\right)^{\ell+1} \int_C d\sigma\, dv d\rho \frac{1}{\Phi_{10}\left(\sigma, v, \rho\right)}\, e^{-i\pi(\sigma n + 2v\ell + \rho m)}$$

The Microscopic Interpretation for $\mathcal{N} = 4$

$$d\left(m, n, \ell\right) = \left(-1\right)^{\ell+1} \int_C d\sigma\, dv d\rho \frac{1}{\Phi_{10}\left(\sigma, v, \rho\right)}\, e^{-i\pi(\sigma n + 2v\ell + \rho m)}$$

- *'Rademacher expansion of a Siegel modular form for $\mathcal{N} = 4$ counting'* - Lopes Cardoso, et al.
- Result in terms of mock modular forms, multiplier systems, Kloosterman sums and Eichler integrals

Future Work

The Macroscopic Interpretation

Future Work

The Macroscopic Interpretation

- *'The gravitational path integral for $N = 4$ BPS black holes from black hole microstate counting'*- Lopes Cardoso, et al.

Thank You

Back

$$K_c(\Delta) := e^{\frac{5\pi i}{4}} \sum_{\substack{-c\leq d<0 \\ (d,c)=1}} e^{2\pi \frac{d}{c}\left(\frac{\Delta}{4}\right)} M\left(\gamma_{c,d}\right)^{-1}_{\nu 1} e^{2\pi i \frac{a}{c}\left(-\frac{1}{4}\right)} \qquad \nu = \Delta \mathrm{mod}\ 2$$

$$M^{-1}\left(\gamma_{c,d}\right)_{\nu 1} = e^{\frac{\pi i}{4}} \frac{1}{\sqrt{6c}} e^{-\frac{i\pi}{6}\Phi(\gamma)} \sum_{\epsilon=\pm 1} \sum_{n=0}^{c-1} \epsilon \exp\left[\frac{i\pi}{6c}\left(d(\nu+1)^2 - 4(\nu+1)(3n+\epsilon) + 4a(3n+\epsilon)^2\right)\right]$$

Where Φ is the Rademacher function

$$\Phi(\gamma) = \frac{a+d}{c} - 12\,\mathrm{sign}(c)\, s(a,|c|)$$

for $c > 0$, the Dedekind sum is

$$s(a,c) = \frac{1}{4c} \sum_{i=1}^{c-1} \cot\left(\frac{\pi i}{c}\right) \cot\left(\frac{\pi i a}{c}\right)$$

Appendix B

Bessel Functions of the First Kind

We define the modified Bessel function $\tilde{\mathcal{I}}_p$ with weight ρ

$$\tilde{\mathcal{I}}_\rho(z) = \left(\frac{z}{2}\right)^{-\rho} \mathcal{I}_\rho(z) = \frac{1}{2\pi i}\int_{\epsilon - i\infty}^{\epsilon + i\infty} \frac{d\sigma}{\sigma^{p+1}} \exp\left[\sigma + \frac{z^2}{4\sigma}\right]$$

where $\mathcal{I}_{\rho(z)}$ is the standard bessel function of the first kind, which satisfies the asymptotic expansion for $z \to \infty$

$$\mathcal{I}_\rho = \frac{e^z}{\sqrt{2\pi z}}\left(1 - \frac{\left(4\rho^2-1\right)}{8z} + \frac{\left(4\rho^2-1\right)\left(4\rho^2-3^2\right)}{2!(8z)^3} - \frac{\left(4\rho^2-1\right)\left(4\rho^2-3^2\right)\left(4\rho^2-5^2\right)}{3!(8z)^5} + \dots\right)$$

The Theta function is given by

$$\Theta_{m,\mu}(\tau, z) := \sum_{\substack{r = \mu \bmod 2m, \\ r \in \mathbb{Z}}} q^{\frac{r^2}{4m}} \zeta^r.$$

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